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Research paper

Bridging techniques and applications in sentiment analysis: Approaches, challenges, and emerging trends

Sachin Deori, Tulika Chutia  , Hirok Jyoti Saikia, Anjan Boro, Nomi Baruah

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Highlights

- Comprehensive review of sentiment analysis techniques and methodologies.
- Covers lexicon-based, machine learning, deep learning, and transformer models.
- Discusses challenges such as sarcasm, low-resource languages, and domain adaptation.
- Explores applications in business, healthcare, politics, and public opinion mining.
- Identifies future directions: multimodal data, interpretability, and ethical issues.
- Analyses 330 articles to summarize methodologies, datasets, and real-world applications.

Abstract

Sentiment analysis (SA) has emerged as a significant research area due to the surge in user-generated content on online platforms. It is crucial for making informed decisions in fields such as politics, business, and healthcare, as it helps to gauge the opinions, sentiments, and attitudes reflected in textual data. This work comprehensively explores the essential concepts, methodologies, and applications of sentiment analysis, including lexicon-based techniques, traditional machine learning, and advanced transformer-based models and deep learning innovations. The study addresses key challenges that affect Sentiment analysis (SA) systems, including sarcasm detection, management of low-resource languages, and adaptability across different domains. It outlines practical applications such as market forecasting, analysis of customer feedback, public opinion tracking, and mental health assessments. Furthermore, the study provides a thorough analysis of 330 research articles detailing core theories, methodologies, and practical implementations of sentiment analysis. It concludes by proposing future research avenues that encompass the use of multimodal data, enhancement of interpretability, and ethical considerations in sentiment analysis.

Introduction

People are increasingly finding that they must communicate online. People may now receive and share information more swiftly thanks to social media's growing acceptability and connectedness. Nowadays, social gatherings and traditional purchase methods have been altered by online

connections (Maitama et al., 2020b, Liu, 2022). Sentiment analysis has been revolutionized by social media since it offers vast, varied, and real-time data. However, the noise and informality of social media information present difficulties that call for sophisticated Artificial Intelligence and Natural Language Processing methods to solve. Businesses, governments, schools, and individuals now use these platforms to communicate with a global audience (Ijairi et al., 2023).

Sentiment analysis is becoming increasingly crucial for researching rapidly expanding viewpoints on social media and other platforms (Aqlan et al., 2019). The importance of sentiment analysis could be seen in different aspects of life, such as business, where consumers' preferences are accessed through online opinion gathering (Maitama et al., 2020b, Qazi et al., 2017). Finding the sentiment polarity – that is, whether a remark is positive, negative, or neutral – is the general justification for sentiment analysis (SA). The Sentiment Analysis must assimilate the data and comprehend its polarity to make the optimal decision (Aqlan et al., 2019).

There are three main classification levels in sentiment analysis: document-level, sentence-level, and aspect-level (Medhat et al., 2014) as shown in Fig. 2.

In document level, it assesses a document as a whole to determine whether it expresses a neutral, positive, or negative sentiment. Usually concentrating on a single primary subject, the document is handled as a single entity. It provides a comprehensive overview of techniques and challenges in this area (Medhat et al., 2014, Behdenna et al., 2016).

Sentence-Level Sentiment Analysis aims to classify each sentence independently to ascertain its sentiment. The first stage in the process is determining whether a sentence is subjective (expressing an opinion) or objective (stating a fact). After further sentence analysis, the sentiment is categorized as neutral, negative, or positive if subjective. This method works well for texts with different feelings between words. It covers techniques for sentiment analysis at the sentence level and is founded on supervised learning and the multi-criteria decision-making method (Medhat et al., 2014, Su et al., 2023).

Aspect-level Sentiment Analysis concentrates on particular elements or characteristics in a sentence or document. It pinpoints feelings associated with specific elements of a topic. "It is Based on Syntax-Aware and Graph Convolutional Networks" investigates sophisticated techniques for assessing sentiments associated with particular textual elements (Medhat et al., 2014, Gu et al., 2024).

The two major Natural Language Processing (NLP) aspects are human language understanding and human language generation (Nandwani and Verma, 2021, Itani et al., 2017). In recent years, deep learning, which is a sub-branch of Artificial Intelligence (AI), has brought many changes to various sectors and contributed to many research projects, and also outperformed humans in tasks related to Natural Language Processing (NLP) (Shome, 2021, Edwards, 2021).

Sentiment Analysis is the process of analyzing digital text to determine the emotional tone of the sentence(positive, negative, neutral), which is used for Natural Language Processing (NLP) (Aqlan et al., 2019, Hussein, 2018).

In the healthcare sector, social media platforms such as X(Twitter) have become an important platform to gather information related to many things shared by doctors, clinical researchers, and the public. One of the examples is the COVID-19 pandemic period, where people around the world have shared their thoughts, feelings, etc on Twitter (Nandwani and Verma, 2021, Garcia and Berton, 2021).

Over the past decades, we have noticed that the rapid growth of user-generated content through Facebook, Twitter, and a lot more on various topics such as political, economic, technology, Artificial Intelligence, etc. That content contains people's feelings, thoughts. Using the social media data, we can analyze and track people's opinions to help predict attitudes towards products, political issues, or even prevent depressed people from committing suicide (Al-Omari et al., 2020, De Choudhury et al., 2013, Bataineh et al., 2019).

Decision-making support: Deciding on any point is an important part of our lives. Information that can be gathered and analyzing that information to make a decision is beneficial, such as "which car to buy", "which place to visit", "means of travelling" (Kaur et al., 2017), etc.

Business application: In the world of business and start-ups, the competition is very tough, and to be the best among other companies, they have to satisfy their clients by taking reviews and, based on the reviews, rolling out products that satisfy customers is a must (Kaur et al., 2017).

Predictions and trend analysis: Sentiment analysis enables us to predict or analyze the updated market by tracking public views. It can be helpful in many sectors, among them is in elections where the representing candidate wants to know the expectations of people from them (Kaur et al., 2017).

- Comprehensive Comparative Framework:

By developing a systematic framework for comparing different sentiment analysis techniques applied to different datasets, the study enables the identification of the most effective tactics in diverse social media scenarios.

- Domain-Specific Insights:

The study illustrates how context affects model performance and sentiment trends by examining sentiment across a variety of domains, including politics, product reviews, and social issues.

- A more structured, repeatable, and context-aware method for assessing sentiment analysis models in practical applications is provided by the research's integration of

the PICOC (Population, Intervention, Comparison, Outcome, Context) framework (see Table 1).

See Table 1.

Despite advancements in lexicon-based, machine learning, and deep learning approaches, there are still a lot of unanswered questions in sentiment analysis. The flexibility of sentiment models is limited because most research ignores low-resource and multilingual scenarios in favor of high-resource languages like English. Additionally, research usually focuses on specific application domains, like political sentiments or product evaluations, while ignoring the cross-domain adaptability of sentiment models. The interpretability and explainability of deep learning classifiers are still problematic, which calls into question transparency and the ethical application of AI. Furthermore, there is no information available regarding the potential for more in-depth sentiment analysis when multimodal data – such as text, voice, images, and video – are integrated. Lastly, it is difficult to find comparison frameworks that systematically evaluate sentiment analysis techniques across a range of datasets and real-life scenarios. Therefore, a comprehensive, domain-aware, and ethically responsible framework is needed to close the methodological, linguistic, and application gaps in sentiment analysis research (see Fig. 1 and Table 2).

This study evaluates many sentiment analysis techniques, such as lexicon-based, machine learning, and deep learning, across a number of areas, including politics, product evaluations, and social concerns, in order to identify research needs. It provides a methodical evaluation process that uses the PICOC framework (Population, Intervention, Comparison, Outcome, Context) to determine the benefits and drawbacks of each strategy in real-world situations. In order to overcome linguistic imbalances and extend the applicability of sentiment analysis beyond high-resource languages, the research focuses on adaptable strategies for multilingual and low-resource contexts. Comparing model performance across multiple datasets offers insights into generalizability and cross-domain adaptability, which are commonly overlooked in previous research. Additionally, the work highlights the importance of explainable and ethical AI by addressing interpretability issues associated with sentiment categorization algorithms. It also identifies future directions for context-aware and morally conscious sentiment analysis frameworks and provides opportunities for integrating multimodal data. The overall objective of the endeavour is to surmount the scientific, linguistic, and ethical obstacles that have hindered the broader application and reliability of sentiment analysis systems.

- **Sentiment Analysis:** The computational process of locating, extracting, and classifying viewpoints expressed in text to ascertain the sentiment polarity (positive, negative, etc.) is known as sentiment analysis (SA).
- **Lexicon-Based Approach:** A sentiment analysis technique that assesses a text's sentiment polarity by using predefined dictionaries of words annotated with sentiment scores.

- **Machine Learning Approach:** A data-driven technique that automatically learns sentiment classification rules by training algorithms on labeled datasets. Random Forests (RF), Naïve Bayes (NB), and Support Vector Machines (SVM) are examples of common algorithms.
- The deep learning approach uses sophisticated neural network-based techniques (such as CNN, RNN, LSTM, and Transformers) that can better classify sentiment by capturing intricate semantic relationships in text.
- **NLP:** The goal of the artificial intelligence subfield of natural language processing (NLP) is to make it possible for computers to comprehend, interpret, and produce human language.

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Section snippets

Literature review

Sentiment analysis has previously been extensively researched using lexicon-based, machine learning, and deep learning techniques – such as SVM, Naïve Bayes, LSTM, and transformer-based models like BERT and XLNet – to classify opinions across domains, such as politics, product reviews, and social media platforms.

Advances in neural architectures and attention-based frameworks have improved contextual understanding, feature extraction, and model performance, according to surveys and studies that ...

Sentiment

The emotional tone or attitude expressed in text, speech, or other forms of communication refers to the word “Sentiment”. It reflects the feelings or emotions of a person, which are positive, negative, or neutral, when the person speaks or writes. It is often used in understanding how people feel or express their opinion about a thing, and that can be any specific topic, product, or related to any event. ...

Emotions

The reactions that human beings experience in response to events or situations are known as ...

Datasets for sentiment analysis

Table 7 lists numerous sentiment and emotion analysis datasets that researchers have used to assess the effectiveness of their models. The Stanford Sentiment Treebank (SST), ISEAR, SemEval, and EmoBank are frequently used datasets for sentiment analysis. These datasets, which include stories, reviews, tweets, and feedback, differ in size, domain, and organization. While EmoBank combines data from news, blogs, and letters and the valence-arousal-dominance (VAD) model, ISEAR concentrates on seven ...

Preprocessing of data

Since people are using social media sites to express their emotions or sentiments by writing text, as a result a massive amount of unstructured data are created on different social media sites and making emotion and sentiment analysis critical for machines to analyze (Chutia and Baruah, 2024). So data pre-processing is a crucial stage in data cleaning process since the quality of the data significantly impacts different approaches that follows the pre-processing steps. ...

Challenges that we identified in existing studies in sentiment analysis

Over the past decade, sentiment analysis has seen a significant amount of work and is still expanding quickly in new ways. But a lot of it is focused on creating sentiment classifiers that are more accurate (Mohammad, 2016). The obstacles of sentiment analysis are numerous and include the presence of linguistic variances, informal writing, and processing cost (Wankhade et al., 2022b). Some of the significant challenges faced in sentiment analysis are: ...

Evaluation metrics

Evaluation metrics are quantitative measures used to assess the performance of a model, algorithm, or system. It is a tool used to assess a model's performance by displaying the counts of true positives, false positives, true negatives, and false negatives (Chutia et al., 2025). It is particularly useful for analyzing sentiment classification. True negatives (TN) indicate the number of non-sarcastic remarks correctly identified by the model, while true positives (TP) represent the number of ...

Sentiment analysis in E-commerce

The future of sentiment analysis (SA) in e-commerce involves several key directions: developing universal models that perform effectively across languages and domains, as most current models are tailored to specific languages or fields (Dashtipour et al., 2020, Yang et al., 2020a, Saleh et al., 2022, Noor et al., 2019, Mabrouk et al., 2021, Hong et al., 2019, Sari et al., 2018, Wongkar and Angdresey, 2019, Huang et al., 2023). Efforts to broaden SA to underrepresented languages like Japanese ...

Challenges and limitations that were identified in this research work

It is essential to discuss the primary challenges and limitations that this systematic review faced to retain openness and direct future research attempts. ...

Conclusion

This review provides a comprehensive understanding of sentiment analysis in the educational domain, focusing on its methodologies, challenges, and future directions. We have emphasized the steps in sentiment analysis, the metrics utilized for evaluation, and the difficulties found in recent research by reviewing the body of existing literature. Lexical ambiguity, managing sarcasm and negation, and the shortcomings of current unsupervised annotation systems in precisely capturing the subtleties ...

Abbreviated terms

The following are the “abbreviated terms” that are used in this study,-

- AI : Artificial Intelligence ...
- BiLSTM : Bidirectional Long Short Term Memory ...
- BERT : Bidirectional Encoder Representations from Transformers ...
- CNN : Convolutional Neural Network ...
- DL : Deep Learning ...
- GRU : Gated Recurrent Unit ...
- ISEAR : International Survey on Emotion Antecedents and Reactions ...
- LR : Logistic Regression ...
- LSTM : Long Short-Term Memory ...
- MAE : Mean Absolute Error ...

- ML : Machine Learning ...
- MSA : Multilingual Sentiment Analysis ...
- MNB : ...

...

CRedit authorship contribution statement

Sachin Deori: Writing – original draft. **Tulika Chutia:** Writing – review & editing. **Hirok Jyoti Saikia:** Writing – original draft. **Anjan Boro:** Writing – original draft. **Nomi Baruah:** Supervision. ...

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